Project: Worked examples
Title: Shaft design
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1. Scope and method

Calculates the bending moment, slope, deflection, fatigue and yield safety factors, critical speed and keys of a stepped shaft on two bearings. It takes radial loads in two planes and a steady torque with an optional alternating part.

WHAT IT CALCULATES	RELATION
Shaft bending moment	the shaft as an S2 beam (pin at A, roller at B) in each plane; $M = \sqrt{(M_y^2 + M_z^2)}$
Stress at a section	$\sigma_a = K_f \cdot 32M_a \cdot d/(\pi(d^4 - d_i^4))$, $\tau_m = K_{fs} \cdot 16T_m \cdot d/(\pi(d^4 - d_i^4))$, $\tau_a = K_{fs} \cdot 16T_a \cdot d/(\pi(d^4 - d_i^4))$; $K_f = 1 + q(K_t - 1)$
Endurance limit	$S_e = k_a \cdot k_b \cdot k_c \cdot k_d \cdot k_e \cdot k_f \cdot S_{e1}$, S_{e1} (the rotating-beam limit S_e') $= 0.5 \cdot S_{ut}$ ($S_{ut} \le 1400$ MPa, else 700 MPa); $k_a = a \cdot S_{ut}$^b, $k_b = 1.24d^{-0.107}$ or $1.51d^{-0.157}$, $k_e = 1 - 0.08 \cdot z_R$
Fatigue (DE-Goodman, bending fully reversed, torque $T_m \pm T_a$)	$1/n_f = \sqrt{(\sigma_a^2 + 3\tau_a^2)}/S_e + \sqrt{3} \cdot \tau_m/S_{ut}$; yield $n_y = S_y/\sqrt{((\sigma_a)^2 + 3\tau_{pk}^2)}$, τ_{pk} from the larger of $T_m + T_a$ and the peak torque T_{pk} when given
Critical speed	Rayleigh $\omega = \sqrt{(g \cdot \Sigma m \cdot y/\Sigma m \cdot y^2)}$, y signed (an overhung mass lifts the span, which lowers the quotient); Dunkerley $1/\omega^2 = \Sigma\, m_i \cdot \delta_{ii}$ (lower bound, used for the ratio to the operating speed)
Weight loads (with w_{masses})	$F_{y,i} = -m_i \cdot g$ at each node, m_i the carried mass plus the shaft's own mass split to the node; without it the loads table is the whole load
Parallel key, on the peak torque T_k = the larger of $T_m + T_a$ and T_{pk}	$\tau = 2T_k/(d \cdot w \cdot L)$, $\sigma = 4T_k/(d \cdot h \cdot L)$; required length $L = \max(2T_k \cdot n_{target}/(d \cdot w \cdot 0.577 \cdot S_{y,key}), 4T_k \cdot n_{target}/(d \cdot h \cdot S_{y,min}))$, $S_{y,min}$ the lowest of the key, the shaft and (when entered) the hub; a shoulder is checked on both sides and the lower n_f governs

Basis: Fatigue, shaft stresses, deflection, critical speeds, keys[1](Ch. 6 (§6-9, Table 6-2 Marin surface factors) and Ch. 7 (§7-4 to §7-7), 10th ed.); Shaft design[2](Ch. 12, 6th ed.).

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
segments	Shaft segments (left to right)	5 row(s)		drawing: GBX-SH-01 rev C	any
loads	Radial loads and masses (a carried mass adds no weight unless w_{masses} is on: enter its weight as a force in F_y)	1 row(s)		spec: Gear mesh forces from the gear calculation	any

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
sections	Sections to check	4 row(s)		assumption: Shoulders r/d ≈ 0.05; keyseat at the pinion (K_t 2.14, K_ts 3.0); q from the Neuber fit.	any
x_A	Bearing A position	20	mm	drawing: GBX-SH-01 rev C	0 … 2000 mm
x_B	Bearing B position	280	mm	drawing: GBX-SH-01 rev C	1 … 2000 mm
S_{ut}	Ultimate tensile strength	700	MPa	spec: EN 10083-3 42CrMo4 +QT, certificate	300 … 1800 MPa
S_y	Yield strength	500	MPa	spec: EN 10083-3 42CrMo4 +QT, certificate	150 … 1900 MPa
E	Young's modulus	207	GPa	assumption: Steel.	50 … 250 GPa
ρ	Density (shaft self-mass in the critical speed)	7850	kg/m³	assumption: Steel.	0 … 12000 kg/m³
T	Torque transmitted, steady or mean part T_m	0.3	kN·m	spec: Motor rated torque × service factor (E1)	0 … 40 kN·m
x_{T1}	Torque enters at	150	mm	drawing: GBX-SH-01 rev C	0 … 2000 mm
x_{T2}	Torque leaves at (optional; blank = the shaft length, so every section beyond x_{T1} carries the torque)	330	mm	drawing: GBX-SH-01 rev C	0 … 2000 mm
w_{masses}	Add the weight of the carried masses and of the shaft itself as downward vertical loads (optional; blank = refused when a mass is carried; off: the loads table is the whole load, the weights are entered as forces and the masses enter the critical speed only)	no		assumption: The loads table is the whole load: the pinion's weight (4 kg, 39 N) is small against the 6 kN mesh force and is not added; the mass enters the critical speed only.	any
n	Operating speed	1450	rpm	spec: Motor nameplate	0 … 20000 rpm
finish	Surface finish	machined or cold-drawn		drawing: GBX-SH-01 rev C	any
R_{rel}	Reliability	99	%	assumption: Design reliability.	90 … 99.9 %

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
k_d	Temperature factor	1		assumption: Below 250 °C.	0.5 … 1.1
k_f	Miscellaneous factor	1		assumption: No other effects.	0.3 … 1.5
n_{target}	Target safety factor	1.5		spec: Company design standard	1.2 … 3
$r_{c,min}$	Smallest ratio of the first critical speed to the operating speed	2		default	1.5 … 4
θ_{max}	Slope limit at the bearings	0.001	rad	assumption: Deep-groove ball bearing slope limit.	0.00001 … 0.1 rad
y_{max}	Deflection limit at the loads	0.13	mm	assumption: Spur gear deflection limit.	0.001 … 10 mm
x_{key}	Key position	150	mm	drawing: GBX-SH-01 rev C	0 … 2000 mm
w	Key width	14	mm	drawing: GBX-SH-01 rev C	1 … 100 mm
h	Key height	9	mm	drawing: GBX-SH-01 rev C	1 … 100 mm
L_{key}	Key length	40	mm	drawing: GBX-SH-01 rev C	1 … 500 mm
$S_{y,key}$	Key yield strength	400	MPa	spec: C45 key steel	150 … 1500 MPa

TABLE ROW	VALUES
segments[1]	x_end = 40 mm; d = 35 mm; d_i = 0 mm
segments[2]	x_end = 120 mm; d = 40 mm; d_i = 0 mm
segments[3]	x_end = 180 mm; d = 45 mm; d_i = 0 mm
segments[4]	x_end = 260 mm; d = 40 mm; d_i = 0 mm
segments[5]	x_end = 340 mm; d = 35 mm; d_i = 0 mm
loads[1]	x = 150 mm; F_y = -2.2 kN; F_z = 6 kN; m = 4 kg
sections[1]	x = 120 mm; K_t = 1.7; K_ts = 1.5; q = 0.8; q_s = 0.9
sections[2]	x = 150 mm; K_t = 2.14; K_ts = 3; q = 0.8; q_s = 0.9
sections[3]	x = 180 mm; K_t = 1.7; K_ts = 1.5; q = 0.8; q_s = 0.9
sections[4]	x = 260 mm; K_t = 1.6; K_ts = 1.4; q = 0.8; q_s = 0.9

3. Assumptions

A1. The shaft is linear elastic on two simple bearing supports, a pin at A and a roller at B. (Shigley 10th ed. §7-4.)
A2. Bending is fully reversed by rotation. (Shigley 10th ed. §7-4.)

A3. The torque is a steady part T_m plus an alternating part T_a, acting between the torque-in and torque-out positions. (The torque is fully alternating when $T_m = 0$.)
A4. The critical speed is the first lateral mode, by Rayleigh with Dunkerley as a lower bound. (Shigley 10th ed. §7-6.)
A5. K_t and K_{ts} come from user records or published fit equations. (No stress-concentration chart is shipped, so the factors are entered or derived.)
A6. At a shoulder both sides are checked, each on its own diameter and bore, and the lower factor governs. (Shigley 10th ed. §6-9, §7-4.)
A7. The load factor k_c is 1, because the stresses are combined by von Mises. (Shigley 10th ed. §6-9, §7-4.)
A8. The loads table is the whole load unless w_{masses} is on, which adds the weights. (With w_{masses} on, the sheet adds m·g at the nodes in the vertical plane.)
A9. Loads, sections and steps closer together than 1/200 of the shaft length share one node. (Keeps the beam model well conditioned.)
A10. sections (Sections to check) is taken as 4 row(s). (Shoulders r/d ≈ 0.05; keyseat at the pinion (K_t 2.14, K_{ts} 3.0); q from the Neuber fit.)
A11. E (Young's modulus) is taken as 207 GPa. (Steel.)
A12. ρ (Density (shaft self-mass in the critical speed)) is taken as 7850 kg/m³. (Steel.)
A13. w_{masses} (Add the weight of the carried masses and of the shaft itself as downward vertical loads (optional; blank = refused when a mass is carried; off: the loads table is the whole load, the weights are entered as forces and the masses enter the critical speed only)) is taken as no. (The loads table is the whole load: the pinion's weight (4 kg, 39 N) is small against the 6 kN mesh force and is not added; the mass enters the critical speed only.)
A14. R_{rel} (Reliability) is taken as 99 %. (Design reliability.)
A15. k_d (Temperature factor) is taken as 1. (Below 250 °C.)
A16. k_f (Miscellaneous factor) is taken as 1. (No other effects.)
A17. θ_{max} (Slope limit at the bearings) is taken as 0.001 rad. (Deep-groove ball bearing slope limit.)
A18. y_{max} (Deflection limit at the loads) is taken as 0.13 mm. (Spur gear deflection limit.)

4. Working

[18] $R_A = \sqrt{(R_{A,y}^2 + R_{A,z}^2)}$
$= \sqrt{(1.1^2 + -3^2)}$
$= 3.1953$ kN
Radial reaction at bearing A
[19] $R_B = \sqrt{(R_{B,y}^2 + R_{B,z}^2)}$
$= \sqrt{(1.1^2 + -3^2)}$
$= 3.1953$ kN
Radial reaction at bearing B
[20] d_A = seat diameter at A
$= x = 20$ mm
$= 35$ mm
Shaft diameter at bearing A
[21] d_B = seat diameter at B
$= x = 280$ mm
$= 35$ mm
Shaft diameter at bearing B
[22] M_{max} = max $\sqrt{(M_y^2 + M_z^2)}$ along the shaft
$= 37$ stations
$= 0.41539$ kN·m
Largest bending moment
[23] θ_A = slope at A (two planes)
$= x = 20$ mm
$= 0.0008961$ rad

Slope at bearing A

[24] θ_B = slope at B (two planes)

= x = 280 mm

= 0.0008961 rad

Slope at bearing B

[25] y_{load} = largest deflection at a load

= 0.071775 mm at 150

= 0.071775 mm

Deflection at the loads

[27] $k_a = a \cdot S_{ut}\text{^}b$ [1](10th ed. §6-9, Table 6-2)

= 4.51 × 700^-0.265 (machined or cold-drawn)

= 0.79474

Surface factor

[28] $k_e = 1 - 0.08 \cdot z_R$ [1](10th ed. §6-9)

= 1 - 0.08 × 2.3263

= 0.81389

Reliability factor

[29] $S_{e1} = 0.5 \cdot S_{ut}$ (S_e')

= 700 MPa

= 350 MPa

Rotating-beam endurance limit

[30] X1 = $\sigma_a = K_f \cdot 32M/(\pi d^3)$, $\tau_m = K_{fs} \cdot 16T_m/(\pi d^3)$, $\tau_a = K_{fs} \cdot 16T_a/(\pi d^3)$; $1/n_f = \sqrt{(\sigma_a^2 + 3\tau_a^2)/S_e}$ + $\sqrt{3}\tau_m/S_{ut}$ [1](10th ed. §7-4)

= x = 120 mm, d = 40 mm, k_b = 0.83561, S_e = 189.17 MPa, K_f = 1.56, K_{fs} = 1.45, M = 319.53 N·m, N·m, σ_a = 79.334 MPa, τ_m = 0 MPa, n_y = 6.3025 (shoulder: the side with d = 40 mm governs, of 2

= 2.3845

Section at 120 mm: fatigue safety factor (DE-Goodman)

[31] X2 = $\sigma_a = K_f \cdot 32M/(\pi d^3)$, $\tau_m = K_{fs} \cdot 16T_m/(\pi d^3)$, $\tau_a = K_{fs} \cdot 16T_a/(\pi d^3)$; $1/n_f = \sqrt{(\sigma_a^2 + 3\tau_a^2)/S_e}$ + $\sqrt{3}\tau_m/S_{ut}$ [1](10th ed. §7-4)

= x = 150 mm, d = 45 mm, k_b = 0.82514, S_e = 186.8 MPa, K_f = 1.912, K_{fs} = 2.8, M = 415.39 N·m, 300 N·m, σ_a = 88.778 MPa, τ_m = 46.947 MPa, n_y = 4.1532

= 1.6909

Section at 150 mm: fatigue safety factor (DE-Goodman)

[32] X3 = $\sigma_a = K_f \cdot 32M/(\pi d^3)$, $\tau_m = K_{fs} \cdot 16T_m/(\pi d^3)$, $\tau_a = K_{fs} \cdot 16T_a/(\pi d^3)$; $1/n_f = \sqrt{(\sigma_a^2 + 3\tau_a^2)/S_e}$ + $\sqrt{3}\tau_m/S_{ut}$ [1](10th ed. §7-4)

= x = 180 mm, d = 40 mm, k_b = 0.83561, S_e = 189.17 MPa, K_f = 1.56, K_{fs} = 1.45, M = 319.53 N·m, 300 N·m, σ_a = 79.334 MPa, τ_m = 34.616 MPa, n_y = 5.0281 (shoulder: the side with d = 40 mm gove of 2)

= 1.9801

Section at 180 mm: fatigue safety factor (DE-Goodman)

[33] X4 = $\sigma_a = K_f \cdot 32M/(\pi d^3)$, $\tau_m = K_{fs} \cdot 16T_m/(\pi d^3)$, $\tau_a = K_{fs} \cdot 16T_a/(\pi d^3)$; $1/n_f = \sqrt{(\sigma_a^2 + 3\tau_a^2)/S_e}$ + $\sqrt{3}\tau_m/S_{ut}$ [1](10th ed. §7-4)

= x = 260 mm, d = 35 mm, k_b = 0.84763, S_e = 191.9 MPa, K_f = 1.48, K_{fs} = 1.36, M = 63.906 N·m, 300 N·m, σ_a = 22.47 MPa, τ_m = 48.465 MPa, n_y = 5.7538 (shoulder: the side with d = 35 mm gover of 2)

= 4.2192

Section at 260 mm: fatigue safety factor (DE-Goodman)

[34] $S_e = k_a \cdot k_b \cdot k_c \cdot k_d \cdot k_e \cdot k_f \cdot S_e$' (governing section)

= k_c = 1

= 186.8 MPa

Endurance limit at the governing section

[35] $n_{f,min}$ = min over sections

= 4 section(s)

= 1.6909

Lowest fatigue safety factor

[36] $n_{y,min}$ = min over sections S_y/σ'_max
= 4 section(s)
= 4.1532

Lowest yield safety factor

[37] $n_c = \sqrt{(g \cdot \Sigma m \cdot y / \Sigma m \cdot y^2)}$ (Rayleigh, y signed) [1](10th ed. §7-6)
= Σm = 7.2337 kg over 37 nodes
= 38275 rpm

First lateral critical speed (Rayleigh)

[38] $n_{c,D} = 1/\omega^2 = \Sigma m_i \cdot \delta_{ii}$ (Dunkerley, lower bound) [1](10th ed. §7-6)
= 37 masses
= 37295 rpm

First critical speed, Dunkerley lower bound

[39] $r_c = n_{c,D}/n$
= 37295 / 1450
= 25.721

Critical-speed ratio (Dunkerley lower bound over operating speed)

[41] $n_{c,req} = r_{c,min} \cdot n$
= 2 × 1450 rpm
= 2900 rpm

Critical speed required, $r_{c,min}$ times the operating speed

[46] $\tau_{key} = 2T/(d \cdot w \cdot L)$ [1](10th ed. §7-7)
= 2 × 300 N·m / (45 × 14 × 40) mm^3
= 23.81 MPa

Key shear stress

[47] $\sigma_{key} = 4T/(d \cdot h \cdot L)$ [1](10th ed. §7-7)
= 4 × 300 N·m / (45 × 9 × 40) mm^3
= 74.074 MPa

Key bearing stress

[49] $L_{key,req} = \max(2T \cdot n_{target}/(d \cdot w \cdot 0.577 S_{y,key}),\ 4T \cdot n_{target}/(d \cdot h \cdot S_{y,min}))$
= n_{target} = 1.5, $S_{y,min}$ = 400 MPa (key 400, shaft 500)
= 11.111 mm

Key length required

[50] $u_{key} = L_{req}/L$
= 11.111 / 40
= 0.27778

Key utilisation

[51] $u_f = n_{target}/n_{f,min}$
= 1.5 / 1.6909
= 0.88712

Fatigue utilisation

[52] $u_y = n_{target}/n_{y,min}$
= 1.5 / 4.1532
= 0.36117

Yield utilisation

[53] $\theta_{bearing} = \max(\theta_A, \theta_B)$
= max(0.0008961, 0.0008961)
= 0.0008961 rad

Larger slope at a bearing

[54] $u_{slope} = \max(\theta_A, \theta_B)/\theta_{max}$
= 0.0008961 / 0.001
= 0.8961

Slope utilisation

[55] $u_{defl} = y_{load} / y_{max}$
= 0.071775 / 0.13 mm
= 0.55212
Deflection utilisation

5. Results

SYMBOL	VALUE	DESCRIPTION
R_A	3.195 kN	Reaction at A
R_B	3.195 kN	Reaction at B
d_A	35 mm	Seat diameter at A
d_B	35 mm	Seat diameter at B
$n_{f,min}$	1.6908	Lowest fatigue safety factor
n_c	38275 rpm	First critical speed (Rayleigh)
M_{max}	0.4154 kN·m	Largest bending moment
θ_A	0.0008961 rad	Slope at A
θ_B	0.0008961 rad	Slope at B
y_{load}	0.07178 mm	Deflection at the loads
S_e	186.8 MPa	Endurance limit
$n_{y,min}$	4.1531	Lowest yield safety factor
$n_{c,D}$	37295 rpm	Dunkerley lower bound
r_c	25.72	Critical-speed ratio
τ_{key}	23.81 MPa	Key shear
σ_{key}	74.07 MPa	Key bearing
$L_{key,req}$	11.11 mm	Key length required
u_{key}	0.28	Key utilisation
u_{crit}	0.08	Utilisation of critical speed against the operating speed
u_f	0.89	Fatigue utilisation
u_y	0.36	Yield utilisation
u_{slope}	0.90	Slope utilisation (governing)
u_{defl}	0.55	Deflection utilisation

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Slope at the bearings against limit	0.0008961 rad	0.001000 rad	0.90	
PASS	Fatigue safety factor against target (DE-Goodman)	1.5	1.6909	0.89	[1](10th ed. §7-4)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Deflection at the loads against limit	0.07178 mm	0.1300 mm	0.55	
PASS	Yield safety factor against target	1.5	4.1532	0.36	
PASS	Key length against required	11.11 mm	40 mm	0.28	
PASS	Critical speed against the operating speed	2900 rpm	37295 rpm	0.08	[1](10th ed. §7-6)

7. Applicability

WARNING records: tabulated values not taken from a reference record are S_{ut}, S_y, E (Shaft material); θ_{max}, y_{max} (Slope and deflection limits). Enter each as one so its source, edition and licence travel with it. [record]
WARNING x_{T2}: the torque acts between x_{T1} = 150 and x_{T2} = 330 mm only. The sections at 120 mm carry no torque, so set x_{T2} if it is carried there. [torque-span]
WARNING $S_{y,hub}$: no hub strength is entered, so the key bearing is checked against the key and the shaft only. The hub is often the weakest part, so enter its yield strength. [key-hub]

Defaults used (inputs left empty):
$r_{c,min}$: not entered, so the default of 2 is used.

8. Verification

```
Template mech.shaft@1.7.0 Content hash 269fa49d39e54f67 Engine calc-engine/1.0.0
Known-answer cases passed 3/3 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
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9. References

1. Budynas RG, Nisbett JK. Shigley's Mechanical Engineering Design. 11th ed. McGraw-Hill Education; 2020.
2. Mott, Machine Elements in Mechanical Design.